D4 - Stockpile Block Model and Model Outputs

Pit I HG stockpile - current state after the 2026-03-29 survey (provisional through 2026-03-31)

1. Model files

File	Format	Use
`outputs/blockmodel/stockpile_block_model.csv`	CSV, one row per block containing material (2,695 blocks)	Excel, Vulcan, Datamine, Leapfrog, Deswik (import with centroid + size)
`outputs/blockmodel/stockpile_block_model.vtk`	VTK structured points, cell data	ParaView 3-D viewing (*Threshold* `tonnes > 0`)
`outputs/blockmodel/block_model_definition.json`	JSON	Origin, block size, dimensions, units, CRS note
`outputs/movement/*.asc`	ESRI ASCII grids (1 m)	Floor plane, pre-record surface, final model surface, latest surveyed surface (GIS / CAD)
`outputs/movement/load_destinations.csv`	CSV	Where each truck load was placed (destination centroid, push distance, spread)
`outputs/blockmodel/summary_*.csv`	CSV	Inventory by material source (**exact** class split), by dominant information class (whole blocks), confidence class, 4 m bench; grade-tonnage

Block definition. Origin (lower-left-bottom corner) E 511,290, N 7,122,280, Z 4,474.0. Blocks 5 × 5 × 2 m; 27 × 26 × 10 blocks. Partial blocks are exported with `fill_frac` and their true material volume. Coordinates are in the mine grid (see D2, A-01).

Units. Tonnes are as reported by the FMS (moisture basis not stated, A-26). Cu is in %, Au and Ag in g/t (as supplied, see QA-F04). Contained Cu is in tonnes, Au and Ag in kg (1 koz = 31.1035 kg).

1.1 Block fields

Field	Meaning	Info type
`ix, iy, iz, xc, yc, zc, dx, dy, dz`	Block index, centroid, size	-
`volume_m3`, `fill_frac`	Material volume in block; fraction of block filled	Measured geometry + inferred where unsurveyed
`tonnes`, `density`	Total tonnes; derived bulk density	Measured / assumed by class
`cu_pct, au_gpt, ag_gpt`	Best-estimate grade (all classes)	Mixed: see class fractions

Field	Meaning	Info type
`cu_t, au_kg, ag_kg`	Contained metal	Mixed
`t_fms, t_unattributed, t_legacy, t_provisional`	Tonnes by information class	-
`frac_fms, frac_unattributed, frac_legacy`	Class fractions	-
`cu_pct_fms, au_gpt_fms`	Grade of the FMS-tracked part only	Measured inputs, model-generated placement
`cu_sd_within, au_sd_within`	Grade heterogeneity among the loads mixed into the block	Model-generated
`n_eff_loads, n_loads_ge1pct, top_loads`	Effective number of loads; loads ≥ 1% of block; top three cycle IDs with share	Model-generated provenance
`mean_place_date`	Tonnage-weighted placement time of FMS material	Measured + model
`top_surveyed_frac`	Fraction of block footprint whose top surface was surveyed	Measured
`cu_p10/p50/p90, au_p10/p50/p90, tonnes_p10/p90, cu_cv, au_cv, tonnes_cv`	Uncertainty (Monte Carlo, 60 realisations; prior load-scale range for blocks < 50% FMS-tracked, see `grade_range_basis`)	Model-generated
`info_class`	Dominant class: FMS-reconciled / Unattributed-imputed / Legacy-assumed / Provisional	-
`confidence`	High / Medium / Low / Uninformed (rules in D3, Section 5.1)	-

2. Inventory

2.1 By information class (base case)

Inventory by information class (FMS grades/metal shown for the whole FMS class)

class	tonnes	share	Cu %	Au g/t	Cu (t)	Au (koz)	basis of grade
FMS-tracked, survey-reconciled	44,422	22.0%	8.42	8.00	3,760	11.5	measured inputs; location model-generated
FMS-tracked, provisional (after last survey)	228	0.1%					measured inputs; location from forward model
Unattributed (surveyed volume, no FMS record)	41,758	20.7%	8.62	8.00	3,598	10.7	imputed (same/adjacent periods)

class	tonnes	share	Cu %	Au g/t	Cu (t)	Au (koz)	basis of grade
Legacy (pre-record pile)	115,699	57.2%	8.42	8.00	9,744	29.8	assumed (FMS mean)
Total modelled	202,107	100.0%	8.46	8.00	17,102	52.0	

> **Two class tables.** This table is the *exact* split: every block's tonnes are divided among the classes. `summary_by_dominant_info_class.csv` instead counts each whole block under its dominant class, for maps and reclaim planning, so its class totals differ even though the grand total is the same.
>
> **Identical grades across classes** follow from the assumptions: legacy grade = FMS mean (A-22), and unattributed grade is imputed from the same loads (A-21).
>
> **Grade units are as supplied and unconfirmed (QA-F04).**

> **Reading this table.** Only the first two rows rest on measured grade data. The grades of the unattributed and legacy rows are imputed or assumed, and their *volumes* depend on assumptions A-13, A-13b and A-14. They must not be summed with the FMS-tracked inventory without that caveat.

2.2 By confidence class

Table 8.1 - Inventory by confidence class

confidence	blocks	tonnes	cu_pct	au_gpt	cu_t	au_koz
High	172	8,971	8.62	8.18	773	2.4
Medium	478	34,468	8.29	7.90	2,857	8.8
Low	412	33,534	8.40	7.95	2,818	8.6
Uninformed	1,633	125,134	8.51	8.03	10,654	32.3

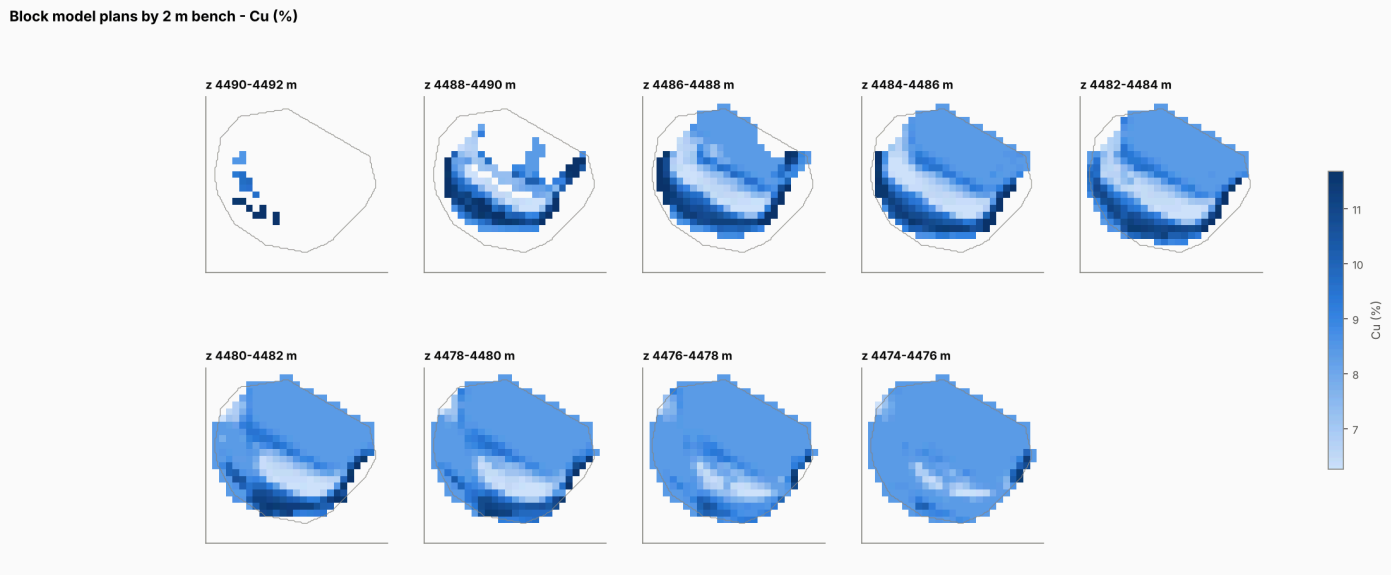
2.3 By 4 m reclaim bench

Table 8.2 - Inventory by 4 m reclaim bench (all classes; see D4 for class split)

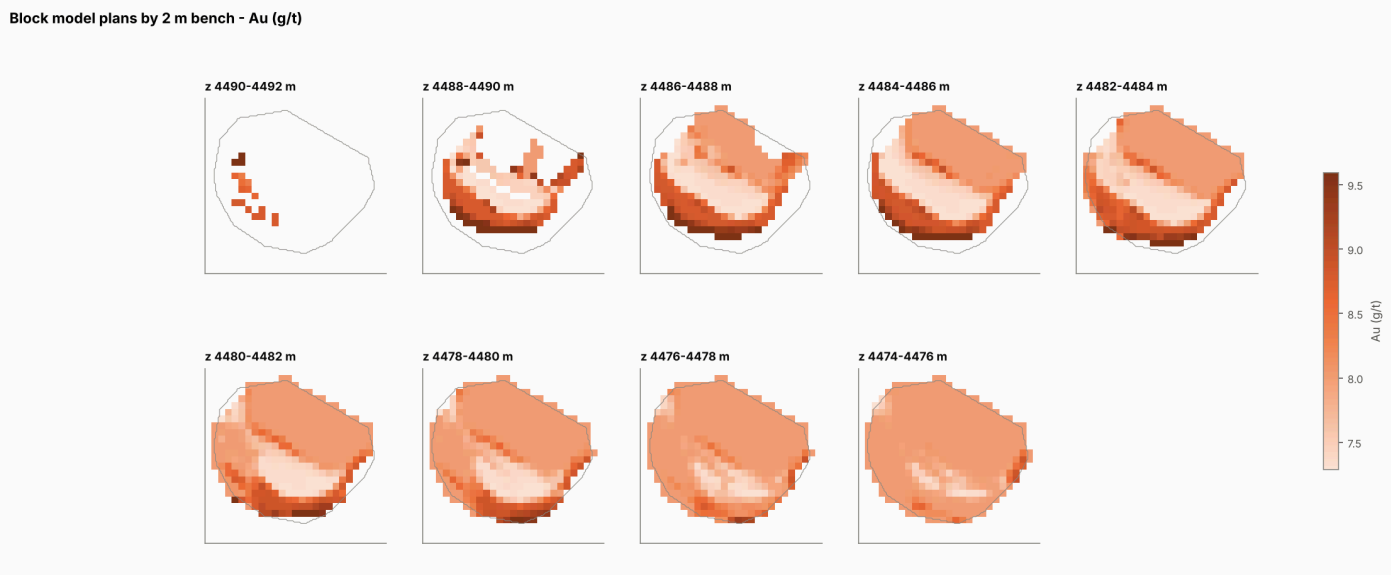
bench	blocks	tonnes	cu_pct	au_gpt	cu_t	au_koz
4490-4494	18	51	9.22	9.59	5	0.0
4486-4490	459	20,958	8.76	8.10	1,836	5.5
4482-4486	671	59,209	8.46	8.00	5,011	15.2
4478-4482	755	68,506	8.46	7.99	5,794	17.6
4474-4478	792	53,383	8.35	7.97	4,455	13.7

Source: `outputs/blockmodel/summary_by_bench_4m.csv` (model run 20261007T164408).

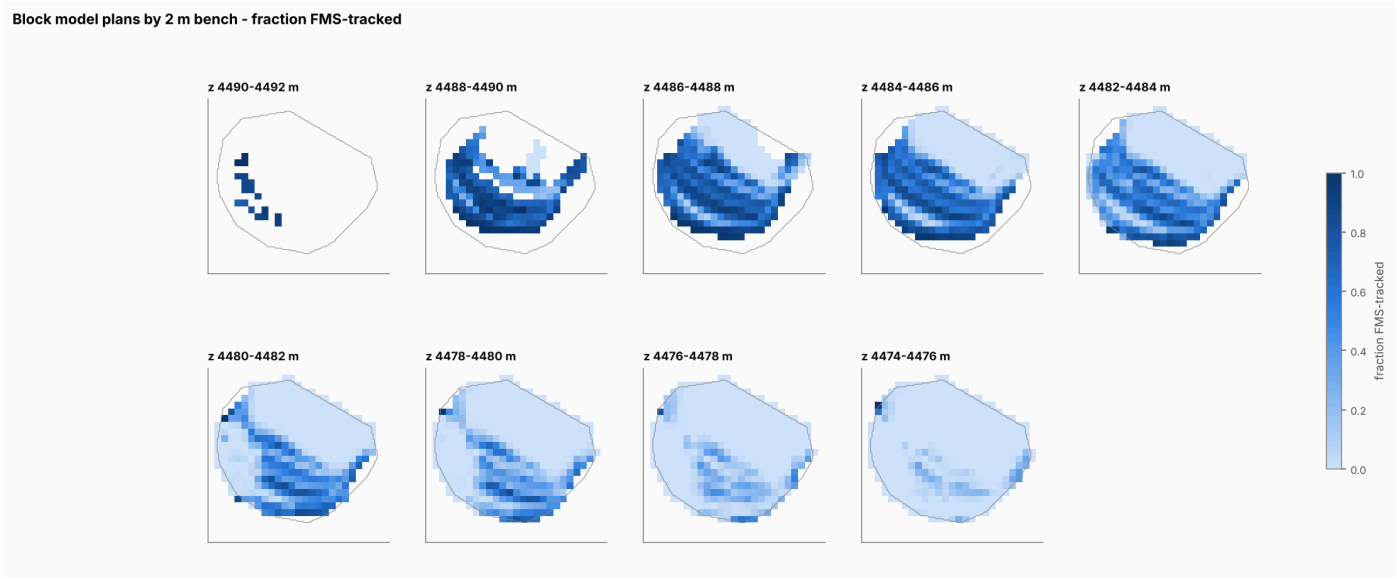
3. Visualisations



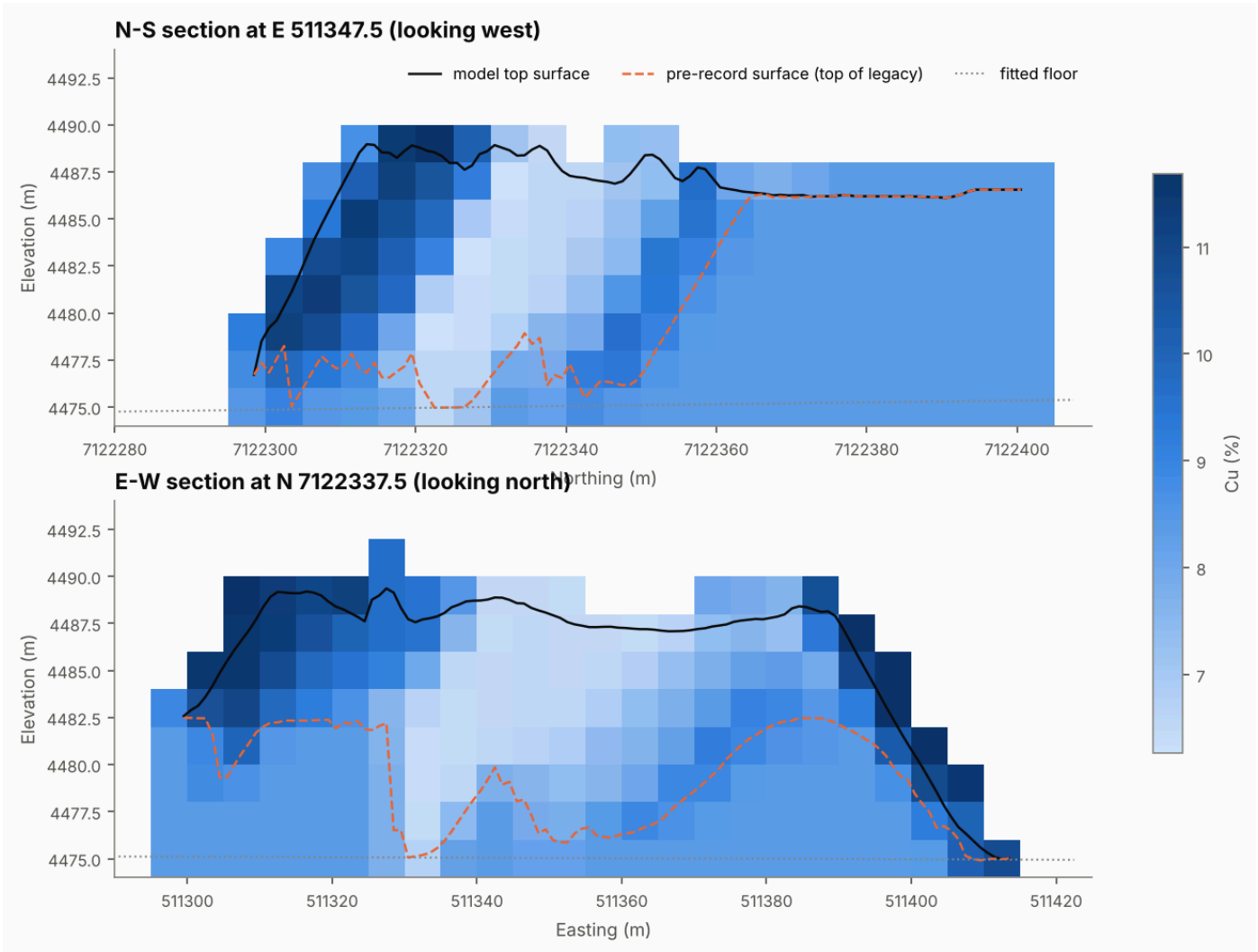
Cu grade by 2 m bench



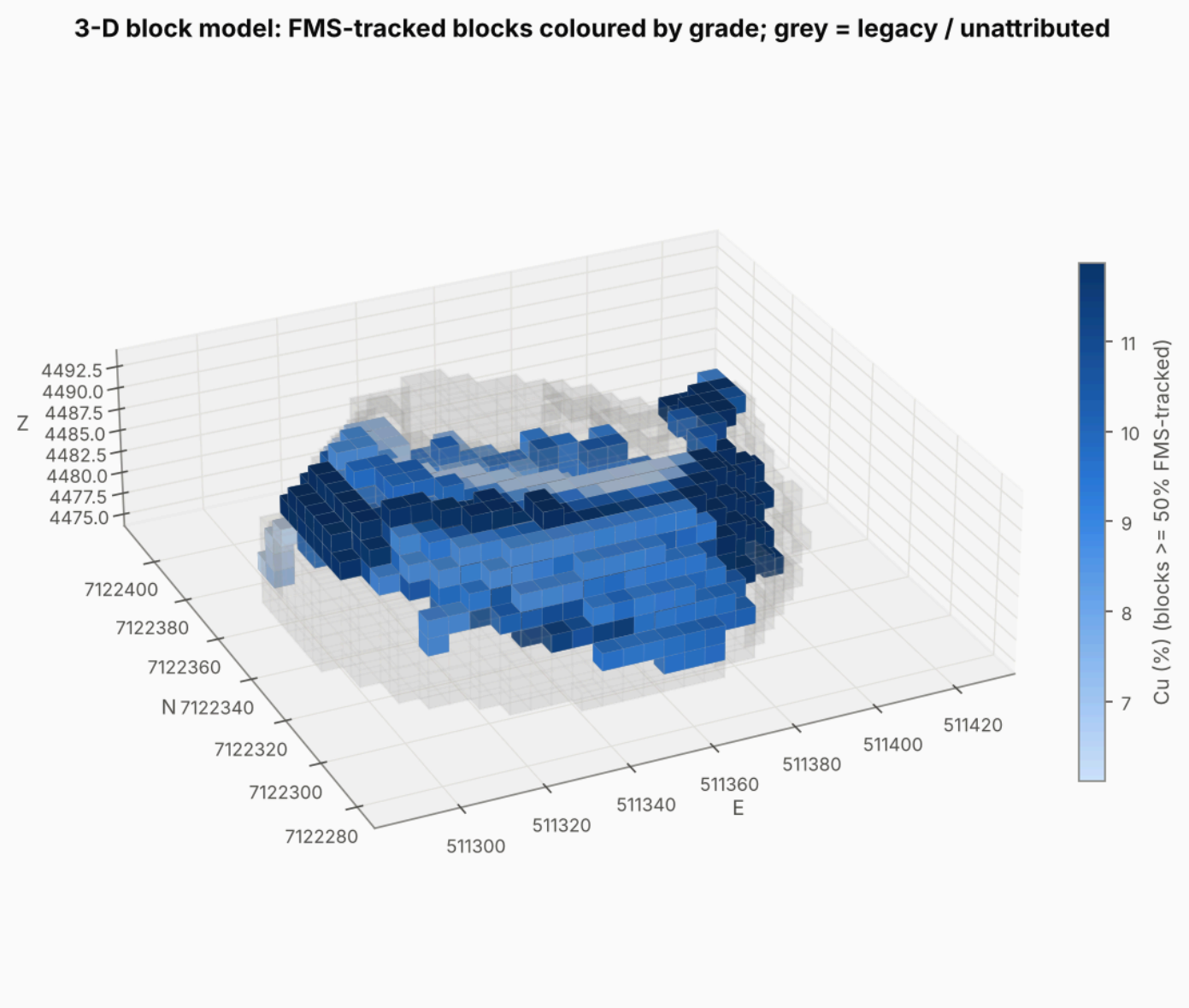
Au grade by 2 m bench



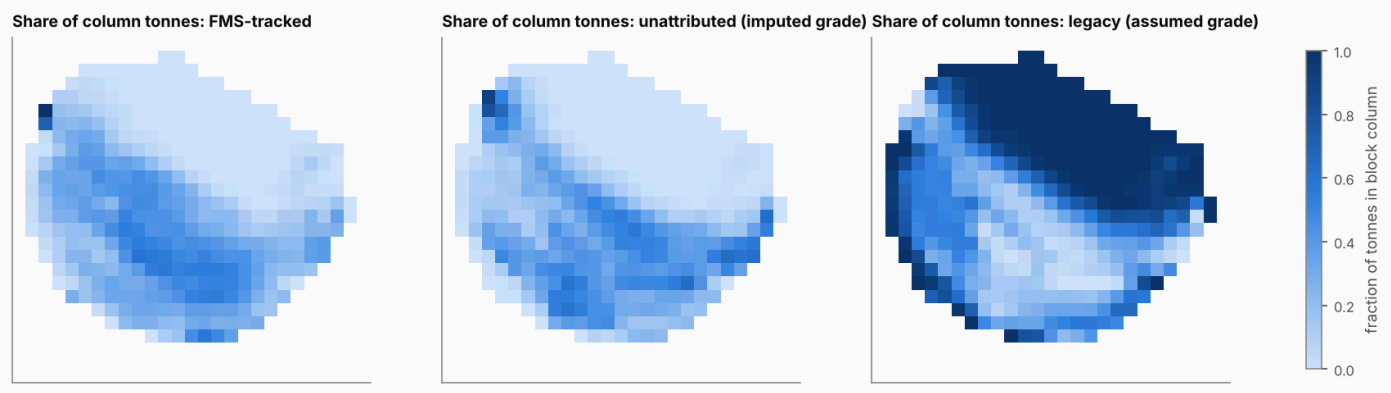
Fraction of block tonnes that is FMS-tracked



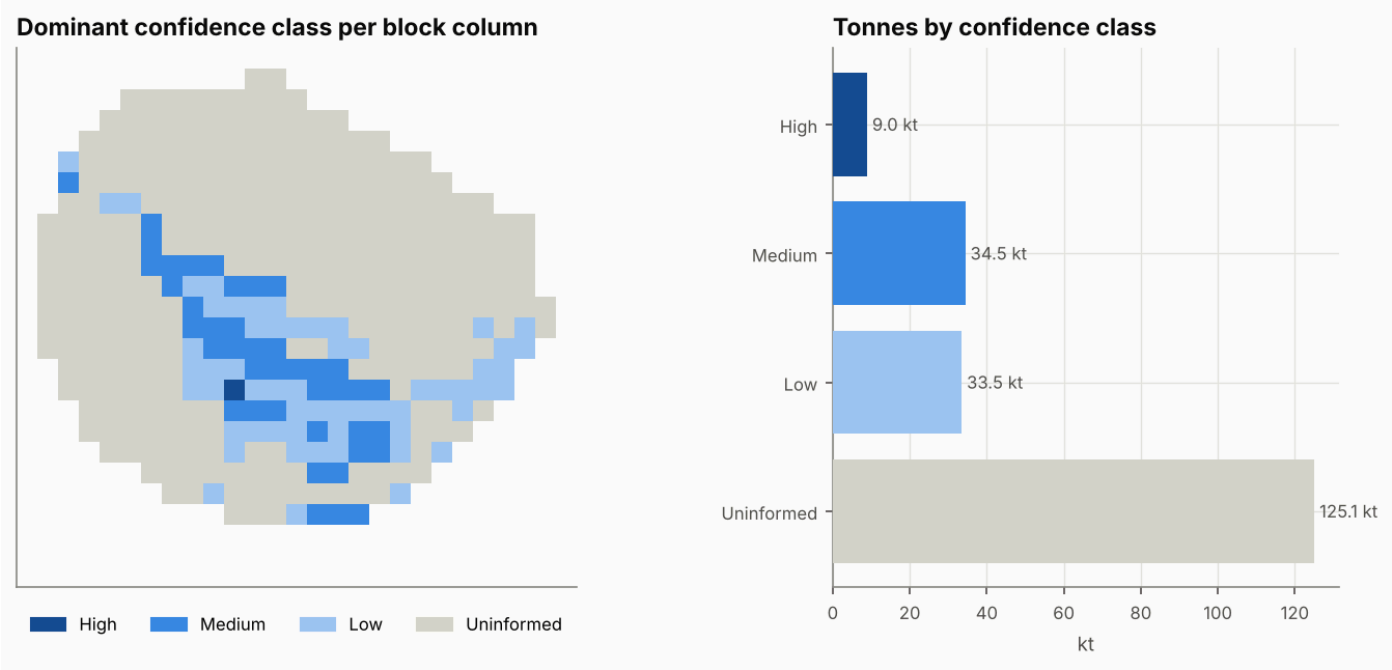
Sections through the tonnage-weighted centre of FMS material



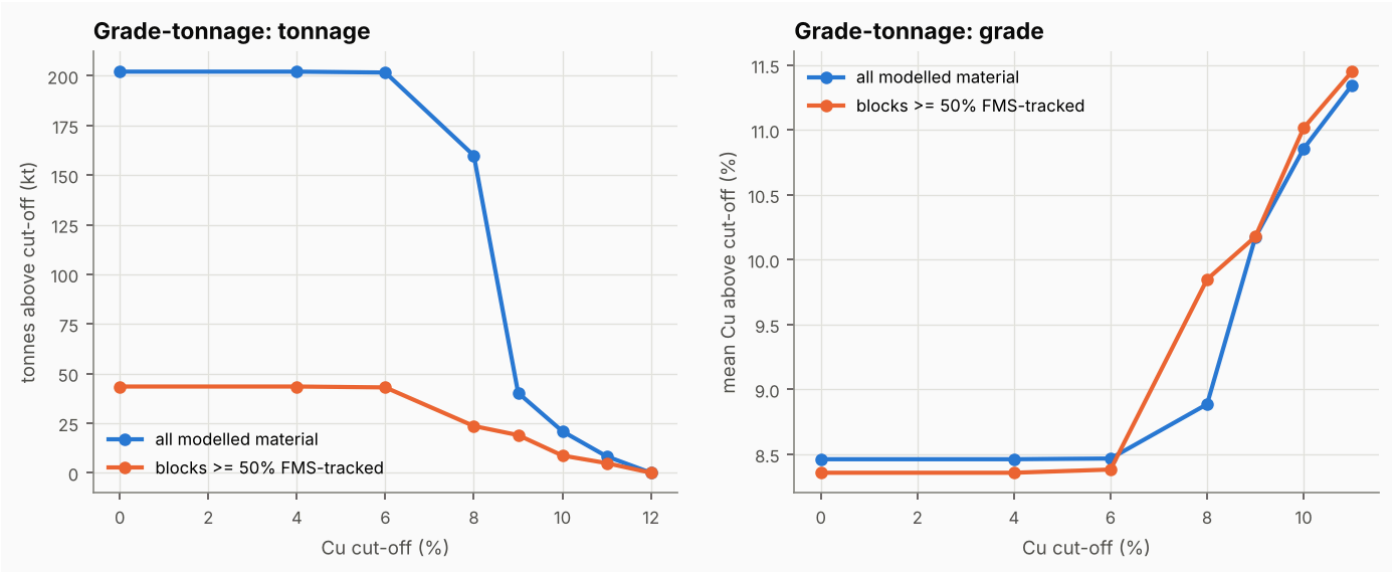
3-D view: FMS-tracked blocks coloured by Cu



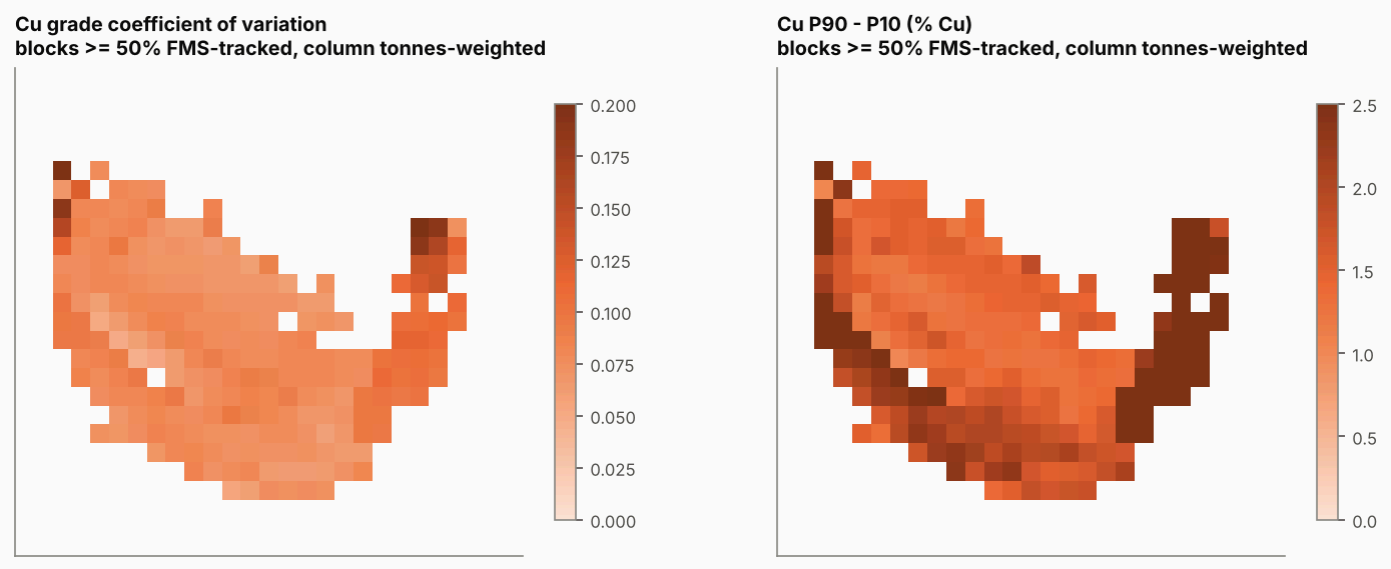
Share of column tonnes by information class



Confidence classes



Grade-tonnage curves



Block uncertainty (Monte Carlo)

4. How to use the model

- **Reclaim sequencing / blending.** Filter `confidence ∈ {High, Medium}` for material whose grade rests on FMS data. Use `cu_p10` and `cu_p90` to test blends against plant constraints. Treat `Legacy-assumed` blocks as unknown-grade material until sampled.
- **Plant-feed / metal forecasting.** Use the bench summaries with their P10-P90 ranges. Report the FMS-tracked inventory separately from imputed and assumed inventories (D6, Section 13).
- **Reconciliation.** `top_loads` and `load_destinations.csv` let mill or stacker grade measurements be compared with the loads predicted to be in a reclaimed block. This is how ε and the push parameters should be calibrated in operation.
- **Updating.** After each new survey run `python -m spbm update`. Provisional blocks will be re-reconciled.